

The AARP logo is displayed in white, bold, sans-serif capital letters. It is positioned in the upper left corner of the slide, partially overlapping the background illustration. The background illustration itself is a vibrant, stylized crowd of diverse people, rendered in a flat, modern art style. The color palette is dominated by warm tones: various shades of red, orange, and yellow, with some cooler tones like purple and grey. The figures are shown in profile or three-quarter view, creating a sense of movement and community. The overall composition is dense and layered, with figures of different ages and ethnicities intermingled.

AARP

Final Status Report

Team 1

Status Report 3 Summary

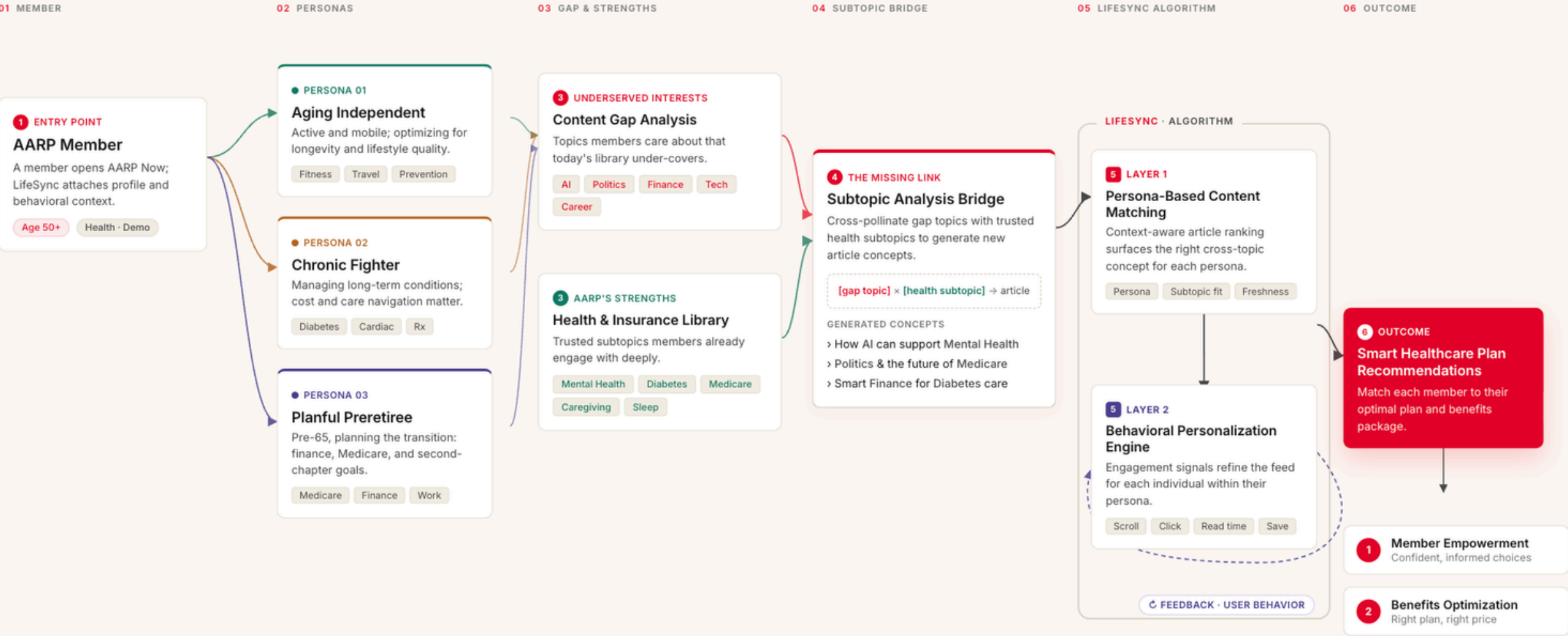


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From a single member to the right healthcare plan, in five steps.

A multi-stage analytical pipeline that segments members into three personas from RAND HRS, Pew Research, and CDC BRFSS using KMeans clustering; quantifies editorial blind spots through bidirectional cosine similarity, BERTopic topic modeling, composite importance scoring, temporal lag analysis, intra-topic semantic spread, UMAP semantic territory mapping, and cross-source centroid alignment, all run on the live AARP and NYT corpora inside the enclave; deepens the gap picture through subtopic clustering with TF-IDF severity scoring and long-term trend analysis to surface unaddressed sub-themes like public health and mental health crisis; validates demand signals externally via Google Trends and zero-shot NLP classification across 80 health policy publications; bridges persona health profiles to AARP's trusted editorial strengths through subtopic semantic analysis; and feeds two algorithm layers, persona-level content matching and real-time behavioral personalization, culminating in plan recommendations scored against the CMS Landscape dataset, driving empowerment, engagement, and revenue.



Data Preparation

Four data sources, cleaned, encoded, and joined.

Article text and metadata from RSS / NYT API; survey data from RAND HRS. KNN imputation was applied only to the HRS persona file , on numeric chronic-condition columns → a clarification added in SR3.

STEP 01 Acquisition

- NYT Developer API
- RSS feeds (titles + metadata)
- RAND HRS extract
- Pew, CDC CDI, Google Trends

STEP 02 Cleaning

- Deduplicate by URL + title
- Normalize timestamps to UTC weekly bins
- **KNN imputation on HRS** chronic-condition columns
- Drop survey rows missing >30% of vars

STEP 03 NLP preprocessing

- Tokenize, lowercase, lemmatize
- Remove stopwords, boilerplate
- Sentence embeddings (MiniLM)
- BERTopic for topic discovery

STEP 04 Feature engineering

- Importance score (vol · recency · consistency)
- Cosine similarity to NYT centroids
- Active-week ratio
- HRS scaled feature matrix for PCA



Literature Review Insight

The trust gap is the strategic problem.

Publicly, AARP advocates for affordable healthcare. Commercially, it earns ~\$1.7B in royalties from UnitedHealthcare. The literature is consistent on how that tension resolves.

North Star

“Earn the right to sell **by first delivering real value.**”

Trust-marketing literature (Edelman 2024, Stengel 2023, Reichheld 2021) converges on the same finding: where commercial relationships are visible, audiences punish concealment and reward useful, audience-first content.

TENSION

\$1.7B/yr in UHC royalties | public advocacy for affordable healthcare.

EVIDENCE

Members already turn to AARP for Medicare and fraud guidance — but not for the broader life decisions of aging.

RESOLUTION

Become a trusted life and health advisor. The royalty relationship becomes the natural next step, not a conflict to manage.

External Data

Four external sources back the analysis. After Sponsor Meeting 2 we moved off direct web scraping → NYT data now comes from the Developer API and RSS feeds for titles and metadata, with no full-article scraping.

SOURCE	USED FOR	RECORDS	ACCESS METHOD	TOS / COMPLIANCE NOTE
RAND HRS (2020 v1)	Persona development	15,204	Public extract	De-identified survey data; no PII used.
NYT Article API Updated	Topic gap, temporal lag	~28,400	Developer API key	Titles & metadata only, no full-article scraping.
AARP RSS Updated	Topic gap, temporal lag	~9,800	Public RSS feeds	RSS for titles & metadata only.
CDC CDI	Persona health profile	aggregate	Public dataset	Aggregate, public domain.
Pew Internet & Tech	Persona digital-use profile	aggregate	Published reports	Cited and linked.
Google Trends	Topic importance signal	weekly	Public Trends UI	Aggregated index, no individual data.

SCQR Framework



Situation	Complication	Question	Resolution
AARP generates most of its revenue through royalties from endorsed partners, most notably UnitedHealth Group’s Medicare and health plan products. To grow these revenues, AARP must continuously drive member engagement with health-related content that builds the trust need for members to act on AARP-endorsed recommendations.	The core challenge is designing an AI platform that adults over 50 want to use. AARP provides trusted, high-quality information, but the experience isn’t yet context-aware, everyone receives the same guidance regardless of personal needs.	A fragmented content strategy weakens the trust seniors need for major decisions and leaves a gap between How can we leverage machine learning to create a hyper-personalized content ecosystem that not only maximizes daily retention but also builds the strategic trust required to naturally funnel members toward AARP-endorsed health and lifestyle products?	We’ll use external data –NYT articles, competitor content, and Google Search Trends—to identify content gaps through EDA, semantic similarity, and topic modeling. Clustering models will define clear customer segments, forming the foundation for an AI-powered life-management platform that strengthens loyalty and links content engagement to AARP’s key revenue streams.

The AARP logo is displayed in white, bold, sans-serif capital letters. A small registered trademark symbol (®) is located to the upper right of the letter 'P'. The logo is positioned in the upper left corner of the image, overlaid on a background of stylized human figures.

AARP®

The background of the entire slide is a vibrant, stylized illustration of a diverse group of people. The figures are depicted in profile or three-quarter view, facing right. They vary in age, ethnicity, and attire, including headscarves and traditional headwraps. The color palette is dominated by warm tones: reds, oranges, yellows, and purples, with some cooler tones like greys and blues for hair and clothing. The style is flat and modern, with clean lines and no facial features other than simple dots for eyes and ears.

Exploratory Data Analysis

Three personas surface from 15,204 HRS respondents.

Each persona has a distinct insurance posture, disease burden, and digital comfort → and a distinct opportunity for AARP.

PERSONA 01

The Aging Independent

34%
of the audience

Insured	14% URGENT
Conditions	1.8 avg.
Digital	Moderate, mobile-first
Tone	Self-reliant, value-conscious

Highest urgency, least protected. Looking for trustworthy, low-pressure plan guidance.

PERSONA 02

The Chronic Fighter

23%
of the audience

Insured	11%
Conditions	3.24 avg. HIGHEST
Digital	Heavy health-content reader
Tone	Pragmatic, info-seeking

Heaviest disease burden. Wants drug pricing, condition management, plan utilization.

PERSONA 03

The Planful Pre-Retiree

43%
of the audience

Insured	41%
Conditions	1.1 avg.
Digital	High; cross-device
Tone	Forward-looking, planful

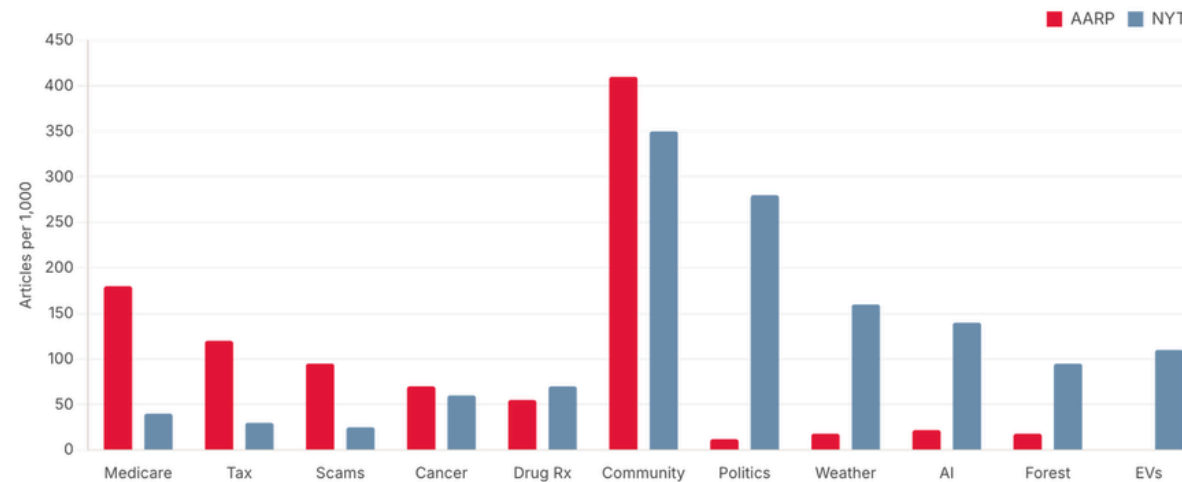
Most receptive to forward guidance — finance, transitions, plan comparison.

EDA confirms the trust and coverage gap.

AARP publishes deeply but narrowly; NYT publishes broadly. **KNN imputation** was applied only to the **HRS persona file**, on numeric chronic-condition columns → not to the article corpus.

ARTICLE VOLUME BY TOPIC • LAST 6 MONTHS

AARP vs. NYT, normalized count per 1,000 articles published.



CORPUS SIZE

28.4k

NYT articles

9.8k

AARP articles

TOPIC COVERAGE

16 topics surface from BERTopic; AARP has zero or near-zero volume on 5.

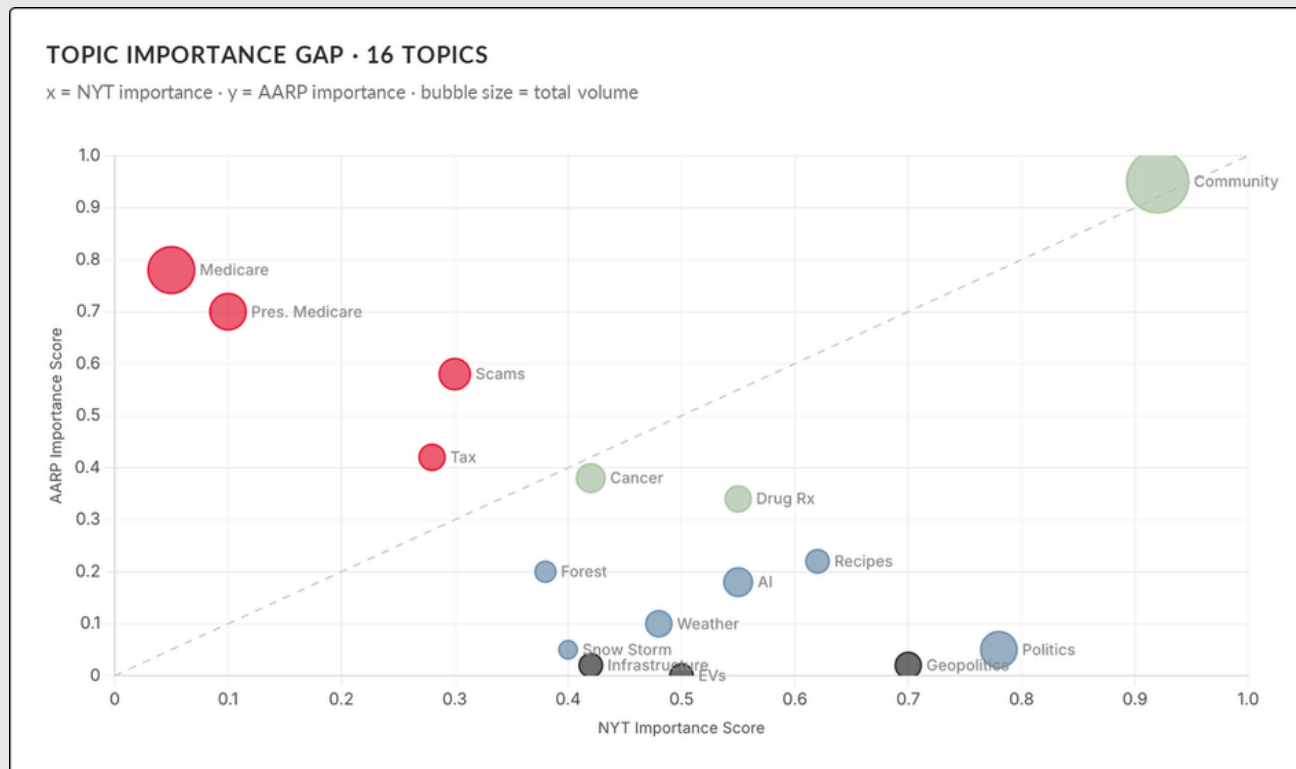
SR3 CLARIFICATION

KNN imputation applied to HRS chronic-condition columns; article corpus uses listwise drop only.

Topic Importance Gap

AARP dominates Medicare; NYT dominates **the broader news cycle**.

Each bubble is a topic; size is total article volume. The dashed line is parity. Above the line, AARP over indexes; below NYT does.



AARP DOMINANT

Medicare, Prescription Medicare, Scams & Fraud, Tax

NYT DOMINANT

Politics, Weather, AI, Forest Science, Recipes

BALANCED

Community, Cancer Care, Drug Prescription

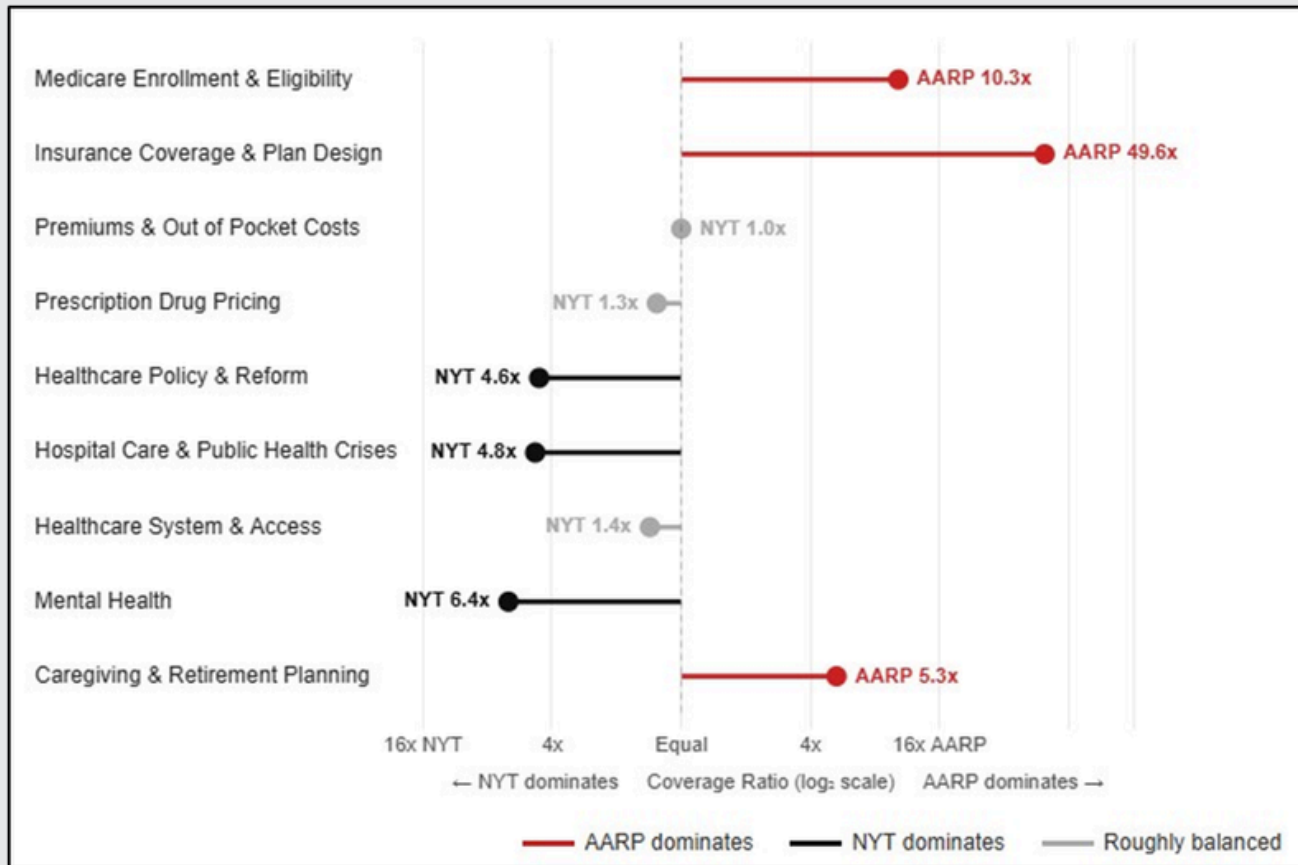
AARP BLIND SPOT

Electric Vehicles · Geopolitics · Infrastructure

Long Term Analysis

Who Dominates Each Healthcare Category?

AARP/ NYT coverage ratio → domain-validated.



AARP DOMINANT

Medicare Enrollment & Eligibility
Insurance Coverage & Plan Design

NYT DOMINANT

Healthcare Policy & Reform
Hospital Care & Public Health
Mental Health

BALANCED

Premiums • Prescription Drug Pricing
Healthcare System & Access

AARP BLIND SPOT

Caregiving & Retirement Planning

NYT-Dominant Areas: sub-topic deep dive

Three categories where NYT Leads → are the hidden gaps worth exploring

Healthcare policy & reform

NYT 4.6x · trend ▼ 34.8%/yr

BLIND SPOT

Public health — NYT: 157 articles · AARP: 0. Highest TF-IDF score across all categories, highest severity score (TF-IDF × volume). Core missed theme.

Trend note

Topic volume declining ~34.8%/yr → market saturating. AARP must find differentiated angles before window closes.

Mental health

NYT 6.4x · trend ▼ 19.9%/yr

WORTH EXPLORING

Mental health crisis — NYT: 27 articles · AARP: 0. Low severity score, but untouched. NYT coverage skewed toward COVID/country-level, an opening for AARP's member-focused lens.

Trend note

Volume declining ~19.9%/yr. Existing coverage is largely COVID-era and time-bound. There may still be potential here, but AARP would need a fresh, member-specific angle (e.g. aging, loneliness) rather than following NYT's approach.

Hospital care & public health crises

NYT 4.8x · no sub-topic data provided

NYT leads substantially in this category. No sub-topic breakdown was provided → flagged for follow-up analysis to identify specific gaps AARP could address.

AARP Blind Spot Intersections

0 AARP articles across all intersections below → sorted by NYT volume

54.9%

NYT articles with ≥1
intersection theme

18.8%

AARP articles with ≥1
intersection theme

0.74

avg themes per NYT
healthcare article

0.20

avg themes per AARP
healthcare article

Blind spot intersection	NYT articles	AARP articles
Hospital Care × AI & Technology	246	0
Hospital Care × Politics & Legislation	80	0
Healthcare System & Access × AI & Technology	64	0
Healthcare Policy & Reform × Global Health	60	0
Mental Health × AI & Technology	60	0
Healthcare Policy & Reform × Politics & Legislation	59	0
Healthcare System & Access × Politics & Legislation	57	0
Hospital Care × Children & Family	46	0
Mental Health × Children & Family	40	0
Healthcare Policy & Reform × Children & Family	33	0

Where AARP Wins & Lags

AARP wins on Medicare and Tax; lags on Politics, Weather, AI, and Forest.

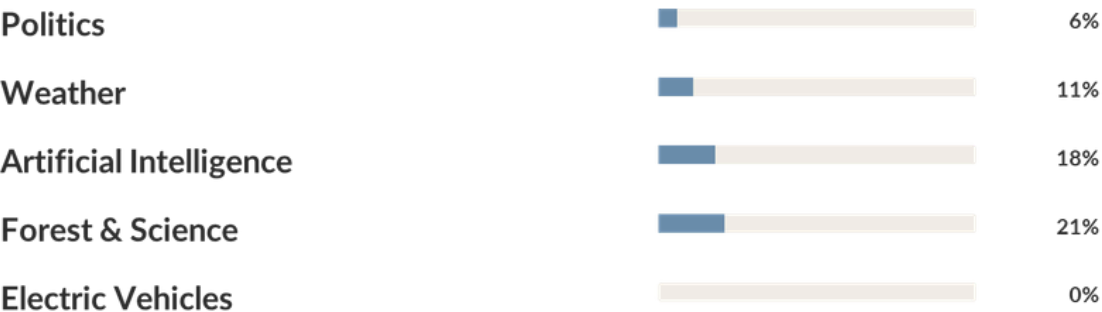
Two-column view of the same 16 topics, sorted by AARP-vs-NYT relative coverage.

WHERE AARP WINS



Strong in core trust topics

WHERE AARP LAGS

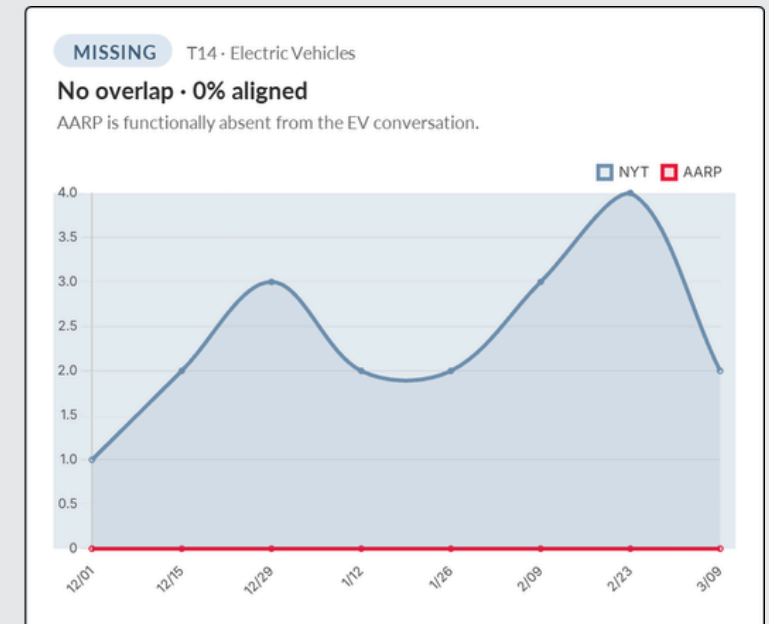
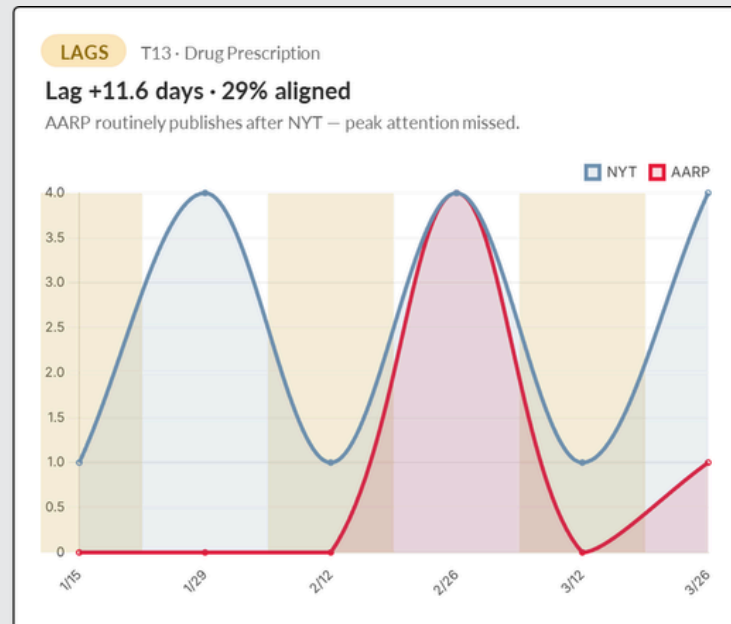
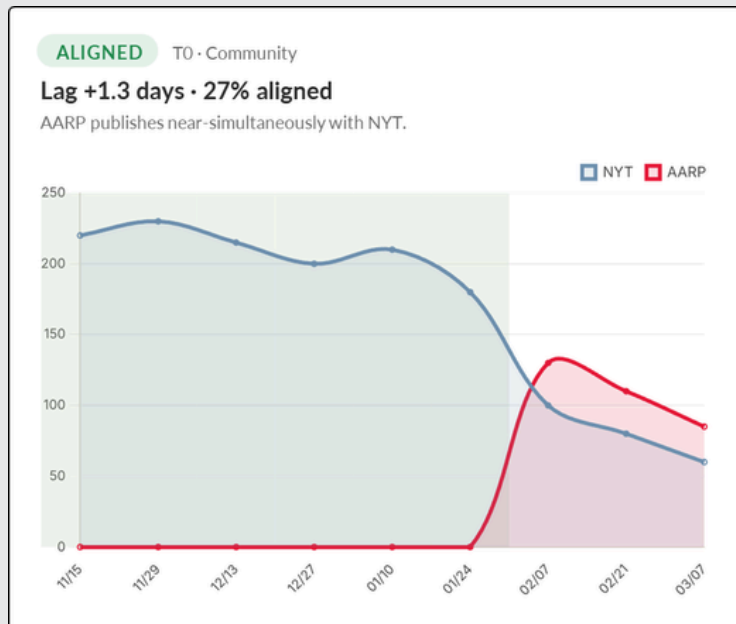


High user attention - low AARP coverage

Temporal Lag

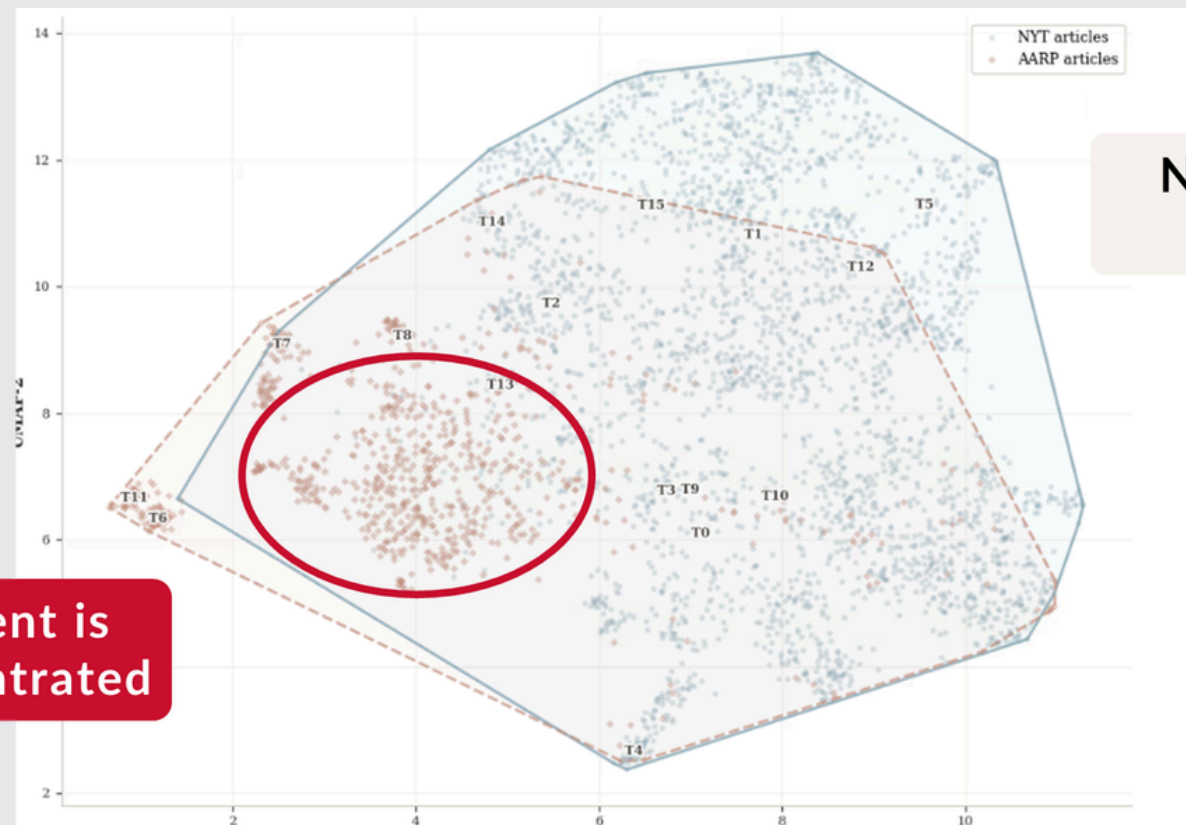
AARP publishes **11.6 days behind NYT** on drug pricing.

Three small multiples → same topics, different temporal posture. Striped bands mark weeks where AARP was silent.



Semantic: AARP shows constrained semantic coverage relative to NYT's broad content landscape

AARP's content is largely embedded within NYT's semantic space, concentrated in health-related domains with focused coverage strategy but limited expansion into broader topic areas.



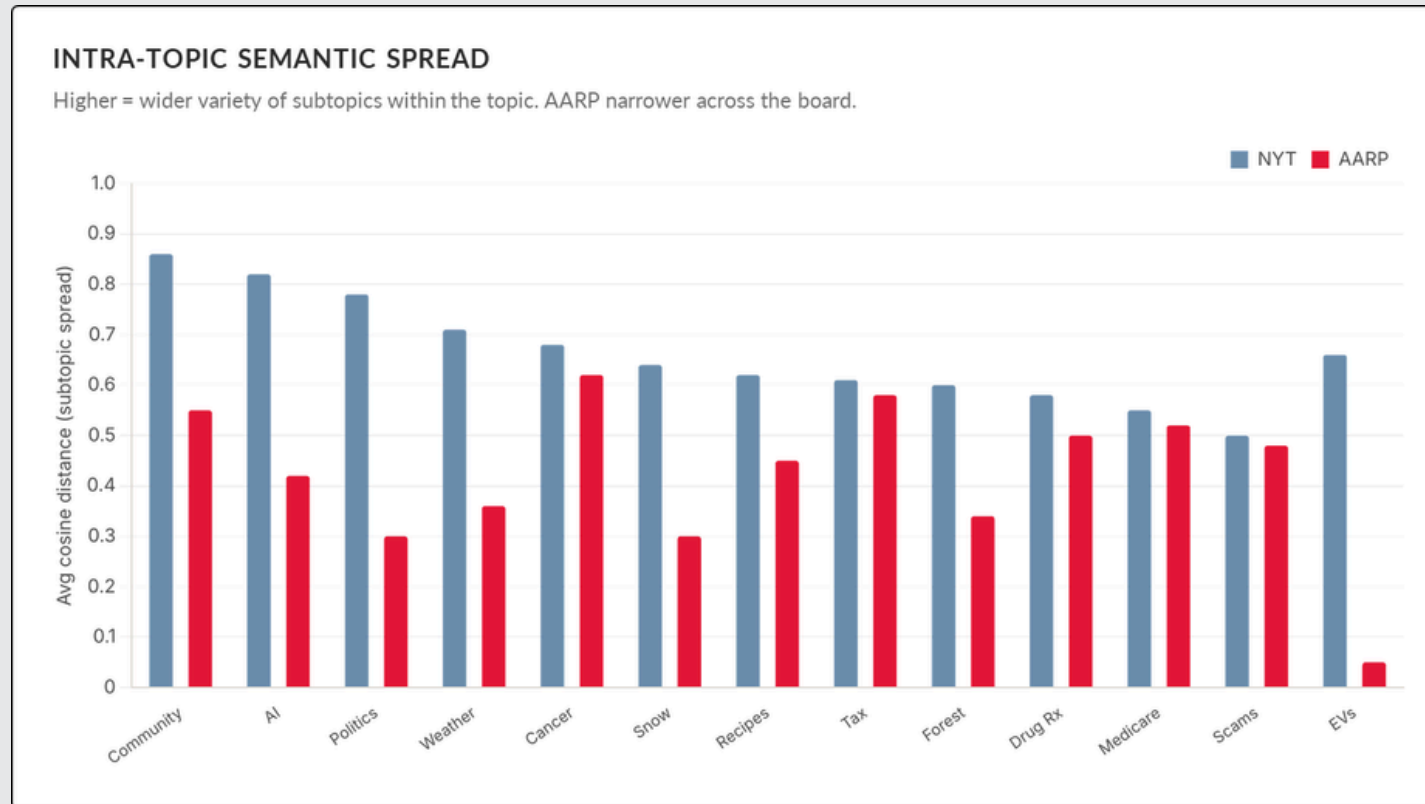
NYT covers a broader range of topics

AARP content is highly concentrated

Subtopic Clustering

Within balanced topics, AARP and NYT cover different angles.

“Subtopic clustering” → formerly “secondary clustering” → shows that even where volumes look balanced, the conversations are different. Volume parity hides angle divergence.



Dive into Healthcare & Insurance:

AARP keeps pace on broad themes but underrepresents emerging and sensitive narratives

Bigram/Tigram Comparative Analysis

Potential Content Gaps found:

1. Insurance coverage incentives
2. Younger Age Death
3. Prescription Data Emergency

Secondary Clustering using HDBSCAN

- Topic 10: No meaningful subtopics
- Topic 13: Three clusters: outliers, drug pricing/insurance coverage, & medical literacy.

Coherence Disparity & Topic Depth Analysis

- AARP's coverage more focused and semantically consistent
- AARP lagged in early-stage coverage of the topic.

Persona x Gap Matrix

Each persona cares about a different **content gap**.

Cell intensity = relative salience of the gap to the persona, derived from HRS demographic + health feautres cross-referenced against topic engagement signals.

CONTENT GAP	Aging Independent	Chronic Fighter	Planful Pre-Retiree	EDITORIAL IMPLICATION
Drug pricing	Low	High	Med	Lead with cost transparency for Chronic Fighter; pair with savings tools for Planful.
Cancer care	Low	High	Low	Condition-management editorial — Chronic Fighter primary.
Plan selection	High	Med	High	Aging Independent + Planful: head-to-head plan comparison.
Politics & policy		Low	Med	Light explainer for Planful; off-strategy for Aging Independent.
AI & tech			Med	Targeted "what AI means for you" series — Planful only.
Weather & climate	Low	Low	Low	Service journalism — health-impact angle.
Electric vehicles			Low	Low priority — defer until other gaps are closed.

LifeSync is **an add-in to AARP Now**, not a replacement

After Sponsor Meeting 2, we repositioned LifeSync. It rides on top of the existing trusted member interface → a personalization layer powered by the persona x gap matrix

LAYER 03

LifeSync New

Persona-curated feed • Health & Wealth Hub

LAYER 02

AARP Now

Existing member interface, content feed, tools

LAYER 01

AARP editorial & member data

38M+ members • existing content library • plan partnerships

WHAT LIFESYNC DOES

Curates the AARP Now feed

By persona + scrolling behavior, pairing gap-analysis topics with AARP's strongest healthcare coverage.

Adds a Health & Wealth Hub

Recommends the right AARP-endorsed Medicare plan and how to maximize it.

WHAT LIFESYNC IS NOT

A standalone app. A replacement for AARP Now. A new editorial brand. The trust comes from AARP itself.

Function 1 → a persona-curated content feed.

AARP brings its existing content into LifeSync. The personalization layer pairs gap-analysis topics with healthcare strengths, scoped per persona.

HOW IT WORKS

Detect persona signal

Scroll, dwell, save — mapped against PCA cluster prototypes.

Pair gap topic + strength topic

A weak coverage area is paired with AARP's strongest related healthcare angle.

Score & surface

Continuous NLP topic scoring keeps the feed responsive to the news cycle.

AARP NOW · LIFESYNC FEED

PERSONA · CHRONIC FIGHTER

DRUG PRICING

GAP TOPIC · PRIORITY 1

What Medicare Part D actually covers in 2026

Paired with AARP's prescription Medicare strength.

CHRONIC CARE

STRENGTH · PRIORITY 2

Living well with multiple conditions: a 6-step routine

High-engagement evergreen from AARP's library.

SCAMS

TRUST ANCHOR

The 5 prescription scams targeting Medicare members this month

AARP-dominant editorial territory.

Function 2 → a Health & Wealth Hub.

Plan selection and plan-utilization guidance, driven by persona, demographics, and engagement signals.

HEALTH & WEALTH HUB

PERSONA • AGING INDEPENDENT

RECOMMENDED FOR YOU

UnitedHealthcare AARP Medicare Supplement • Plan G

MATCH • 92%

PREMIUM • \$148/MO

MAX OUT-OF-POCKET • \$0

WHY THIS FIT

- Self-pay history
- Prefers predictable costs
- 1.8 conditions on average

MAXIMIZE IT

- Wellness benefit • \$200/yr
- Annual review reminders
- Drug-pricing tool linkout

Three alternatives shown below; comparison is a single tap away.

INPUTS

- Persona cluster + sub-cluster
- Self-reported demographics
- Engagement signals from feed
- Geo eligibility & partner availability

OUTPUTS

- Single best-fit recommendation
- 2 alternatives, scored side-by-side
- Utilization tips per plan benefit

TRUST SCAFFOLDING

Always presented after — not before — relevant editorial content. Earn the right to recommend.

Business Impact & Revenue Opportunity

\$16M - \$80M in incremental annual royalties.

Sized against the addressable Medicare-eligible population without supplemental coverage.

ADDRESSABLE MARKET

6.7M

Medicare beneficiaries without a supplemental plan — the population LifeSync is positioned to serve.

CONSERVATIVE · 2%

CONV.

\$16M

/ year incremental

MODERATE · 5%

CONB

\$80M

/ year incremental

Math: $6.7M \times \text{conv.} \times \120 avg. annual royalty per supplemental plan placement.

Sponsor Meeting #2 → Takeaways

Three pivots since Sponsor Meeting 2.

PIVOT 01 · DATA

From scraping to RSS & API

Before Direct web scraping of AARP & NYT

After **NYT Developer API + RSS feeds for titles & metadata**

Resolves ToS concerns raised by sponsor.

PIVOT 02 · NARRATIVE

Content-gap analysis is the deliverable

Before Multiple parallel analyses competing for emphasis

After **Topic gap × persona is the spine; everything else supports it**

Sharper sponsor-facing story.

PIVOT 03 · PRODUCT

LifeSync = add-in, not replacement

Before LifeSync as a standalone product surface

After **Add-in to AARP Now — personalization on a trusted base**

Reduces adoption friction, leverages existing trust.

Recommendations & Next Steps

Close the gaps, ship the add-in, keep the loop running.

REC 01

Close core content gaps
Drug pricing, cancer care,
Medicare supplements —
measurable in 90 days.

REC 02

**Deploy LifeSync inside AARP
Now**
Phase 1: feed curation. Phase
2: Health & Wealth Hub.

REC 03

Finalize persona × gap matrix

Lock cell weights against an
editorial calendar.

REC 04

**Complete healthcare subtopic
analysis**

Plan-recommendation
refinement at sub-condition
resolution.

REC 05

Continuous NLP topic scoring

An ongoing editorial signal —
not a one-time snapshot.

Saturday, May 2

Good morning, Judy.

Here's today's read — a mix of what's happening in the world and what we think you'll find useful.

 3 community events near you this week

✦ Powered by LifeSync

Today's Read

Refreshed this morning · 9 stories



Living Well with Chronic Pain: A Day-by-Day Playbook

Pain physician Dr. James Okafor shares the small daily routines — gentle morning stretches, anti-inflammatory meals, and sleep hygiene tweaks — that members say have changed how they manage flare-ups, without leaning harder on medication.

🕒 7 min read

[Read Article >](#)

Team Member Contribution

April 12th - May 1st

Melody Borg	Yuhan Ding	Everett Ypsilantis	Prathiba Reddy	Scarlett Shi
• Gathering Life-Trigger & Demographic Data • External Data Analysis and Persona Creation • The “LifeSync” Platform Product Strategy	• NLP & Topic Matchmaking • Similarity & Semantic Analysis	• Targeted Marking Strategy • Status Report Construction • The “LifeSync” Platform Product Strategy	• NLP & Topic Matchmaking • Similarity & Semantic Analysis	• Extracting the Core Text Data • NLP & Topic Matchmaking • Similarity & Semantic Analysis



AARP

Thank you again for the opportunity to partner with AARP.